

SECTION B

7 **(short)** A uniform rod of length l and mass m is released from the vertical position shown in Fig. 7 and commences to fall under gravity. The rod rotates clockwise about the pivot point O where θ is the angular position of the rod with respect to the vertical. For the case when $\theta = 45^\circ$ determine:

- (a) the angular acceleration of the rod; [5]
- (b) the angular velocity of the rod. [5]

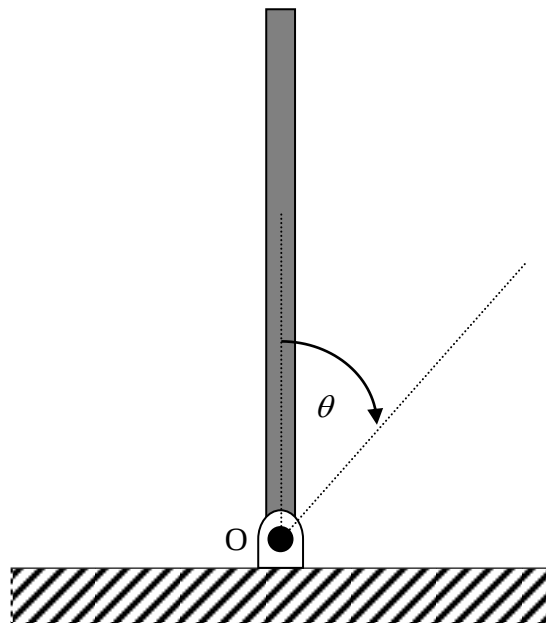


Fig. 7